CST8248 – Emerging Technologies

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Week 04

CST8248

Agenda

- SAN
- Fibre Channel
- iSCSI

NAS vs SAN

- NAS – Network Attached Storage
  - Network attached storage shares the server's host file system. File transactions are managed by the file server. (SMB, NFS, CIFS)
- SAN – Storage Area Network
  - Storage Area Networks share disk blocks. The client system (initiator) must provide a files system for storing files. The client system is also responsible for file transactions (FC, FCoE, iSCSI)

SAN Terminology

- Fibre Channel – FC
- Internet Small Computer System Interface - iSCSI
- Initiator – The “Client” in a SAN. Initiators request storage from targets
- Target - The Server in a SAN . The target checks the WWNN of the initiator and produces the LUN’s appropriate for the initiator

SAN Terminology

- Logical Unit Number – LUN
  - A value assigned to a virtual disk for identification purposes.
- Virtual Disk
  - Also known as a Logical Unit. This is a subdivision of a disk group which will be seen as a block device by the initiator.
- Host Bus Adapter
  - HBA – these devices are used to connect the initiator and target. Switches may be used to modify the connection topology. The connection topology is known as a fabric

Fibre Channel (FC)

- Fibre Channel supports speeds of 1,2,4,8,16,32,64 and 128 Gbps
- Proprietary equipment makes up FC fabrics and can be expensive.
- Dedicated HBA's are used connect with a variety of fiber optic connections
  - Modern systems tend to use LC type connectors
  - Other systems may have used SC type connectors

Fibre Channel

Fibre Channel

- WWN – World Wide Name
  - A unique address identifying devices (HBAs)
- WWNN – World Wide Node Name
  - An Address used to identify a node which can identify a collection of HBAs or Multiport HBAs
- WWPN – World Wide Port Name
  - A unique address to an individual port on Multiport HBAs
- WWN's are 8 Bytes long consisting of 16 Hexadecimal Characters
  - Eg. 20:00:00:00:c9:ca:89:55

iSCSI

- Internet Small Computer Systems interface
- Runs over Ethernet and Ethernet Switches
- Non-Proprietary equipment
- Can operate in conjunction with IP Networks
- Default Port: 3260

iSCSI

- Can use dedicated HBAs
  - TOE – TCP Offload Engine
  - TOE is a hardware feature of NICs that allow information coming through the NIC to be processed by the NIC itself rather than the CPU

iSCSI Addressing

- IQN – iSCSI Qualified Name
  - This is the WWNN or WWPN for an iSCSI network
  - IQN's are unique to the host
- EUI – Extended Unique Identifier
  - Similar to IPv6 EUI can be used to uniquely identify devices in an iSCSI Network.

Example:

iqn.2001-04.com.example:storage:diskarrays-sn-a8675309

RFC 3720

- RFC 3720 Identifies the entire iSCSI standard and details the IQN as follows:

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The iSCSI qualified name string consists of:

      -  The string "iqn.", used to distinguish these names from "eui."
         formatted names.
      -  A date code, in yyyy-mm format.  This date MUST be a date
         during which the naming authority owned the domain name used in
         this format, and SHOULD be the first month in which the domain
         name was owned by this naming authority at 00:01 GMT of the
         first day of the month.  This date code uses the Gregorian
         calendar.  All four digits in the year must be present.  Both
         digits of the month must be present, with January == "01" and
         December == "12".  The dash must be included.
      -  A dot "."
      -  The reversed domain name of the naming authority (person or
         organization) creating this iSCSI name.
      -  An optional, colon (:) prefixed, string within the character
         set and length boundaries that the owner of the domain name
         deems appropriate.  This may contain product types, serial
         numbers, host identifiers, or software keys (e.g., it may
         include colons to separate organization boundaries).  With the
         exception of the colon prefix, the owner of the domain name can
         assign everything after the reversed domain name as desired.
         It is the responsibility of the entity that is the naming
         authority to ensure that the iSCSI names it assigns are
         worldwide unique.  For example, "Example Storage Arrays, Inc.",
         might own the domain name "example.com".
```

https://tools.ietf.org/html/rfc3720

iSCSI

- Multipathing
  - Multiple pathways are often very common in mesh fabric topologies
  - The initiator is responsible for identifying multiple paths to the target storage and selecting one path for use for each session
- Security
  - iSCSI supports IPSEC for end to end encryption
  - CHAP – Password protected connections

iSCSI - Procedure

- Disk Groups – Individual Disks are arranged into collections usually with RAID or similar technology
- Virtual Disks – Segments of a disk group can be made into Virtual Disks. Virtual Disks are assigned a Logical Unit Number which identifies the Virtual disk uniquely to the Storage array.

iSCSI - Procedure

- Mapping virtual disks to hosts. Generally, one LUN is mapped to one host. In our physical datacenter environment one LUN is mapped to a host group consisting of 3 or more hosts
- On the iSCSI Target configure the initiator's information. This is usually the system hostname and associated IQN

iSCSI - Procedure

- On the iSCSI initiator configure the target IQN and IP address
- If CHAP is being used set a CHAP Secret on the target and ensure that the matching CHAP Secret is provided to the initiator

iSCSI Deployment Demo

- Deploy a platform for Target.
- Add Storage
- Manage iSCSI Network interface
- Install iSCSI services
- Deploy a platform for an Initiator
- Connect Target to Initiator
- Connect Initiator to Target

iSCSI Operation Procedures

- SAN protocols are more akin to local storage than they are to NAS protocols
  - SAN protocols offer block devices
  - SAN protocols can operate using the same storage commands a local disks (iSCSI)
- Connection failure over a SAN may lead to file corruption or OS panic (BSOD)
- As such using SAN protocols for client storage is not generally advised

iSCSI Operation Procedures

- Always ensure the Target Server in a SAN is started first and the service is operational prior to starting the initiator
- More importantly: Always shutdown initiators prior to shutting down storage providers (targets).